The most important examples of Lie groups are **linear algebraic groups**, which are subgroups of $GL_n(\mathbb{R})$ defined by algebraic equations (such as, for example, the group of orthogonal matrices $O_n(\mathbb{R})$).

Also, given a Lie subalgebra $\mathfrak{g} \subset \mathfrak{gl}_n(\mathbb{R})$ (which, by Ado's theorem, can be any finite dimensional real Lie algebra), we can define G to be the subgroup of $GL_n(\mathbb{R})$ generated by the elements e^X, $X \in \mathfrak{g}$. One can show that this group has a natural structure of a connected Lie group, whose Lie algebra is $\mathfrak{g}$ (even though it is not always a closed subgroup). While this group is not always simply connected, its universal covering $\widetilde{G}$ is, and it is the Lie group corresponding to $\mathfrak{g}$ under Lie's correspondence.

For more on Lie groups and their relation to Lie algebras, the reader is referred to textbooks on this subject, e.g. [**Ki**].

2.10. Historical interlude: Sophus Lie's trials and transformations

To call Sophus Lie (1842–1899) an overachiever would be an understatement. Scoring first at the 1859 entrance examinations to the University of Christiania (now Oslo) in Norway, he was determined to finish first as well. When problems with his biology class derailed this project, Lie received only the second-highest graduation score. He became depressed, suffered from insomnia, and even contemplated suicide. At that time, he had no desire to become a mathematician. He began working as a mathematics tutor to support himself, read more and more on the subject, and eventually began publishing research papers. He was 26 when he finally decided to devote himself to mathematics.

The Norwegian government realized that the best way to educate their promising scientists was for them to leave Norway, and Lie received a fellowship to travel to Europe. Lie went straight to Berlin, a leading European center of mathematical research, but the mathematics practiced by local stars — Weierstrass and Kronecker — did not impress him. There Lie met young Felix Klein, who eagerly shared this sentiment. The two had a common interest in line geometry and became friends. Klein's and Lie's personalities complemented